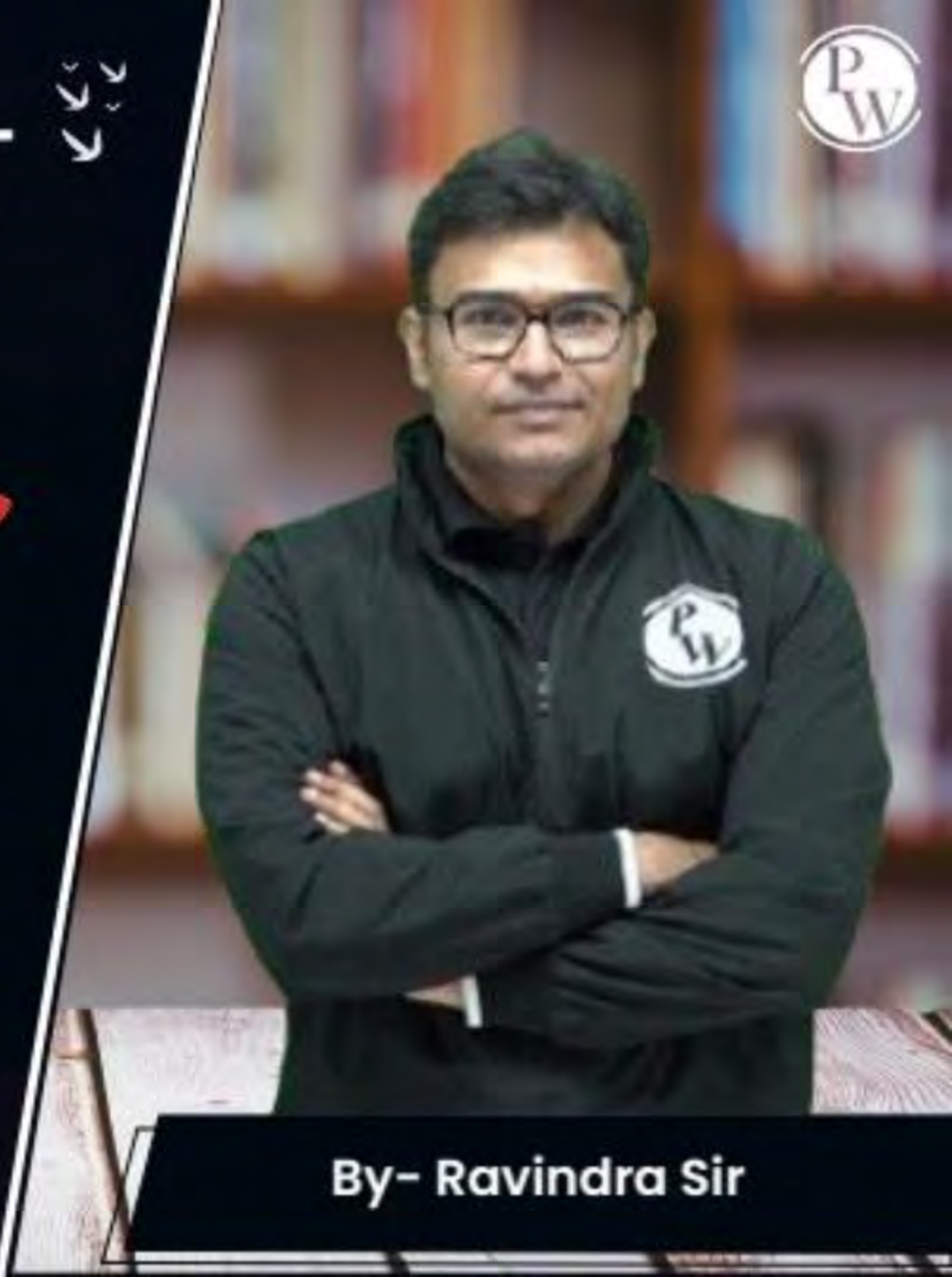
Computer Science & IT

Theory of Computation



regular languages

Lecture No. 9



By- Ravindra Sir

Recap of Previous Lecture



Topic DFA

Topic NFA

Topic RE

Topics to be Covered



Topic DFA –

Topic NFA –

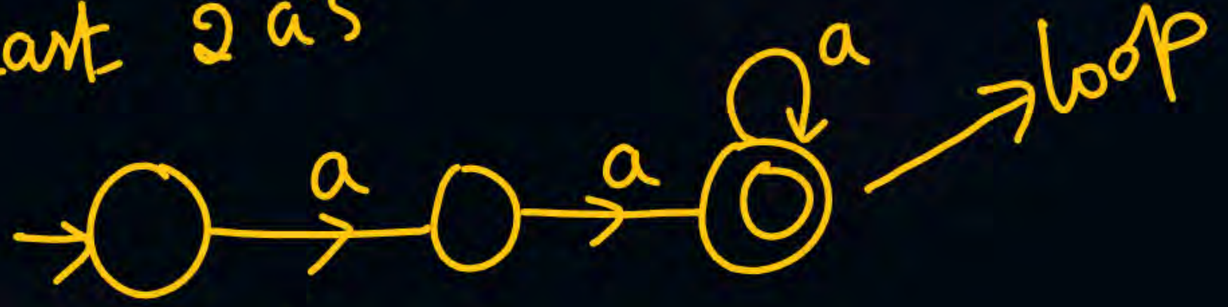
Topic RE



Glid → RE properties → Conversion of FA → RE

Grid :→ fa counting two d mde symbols

atleast 2 a's



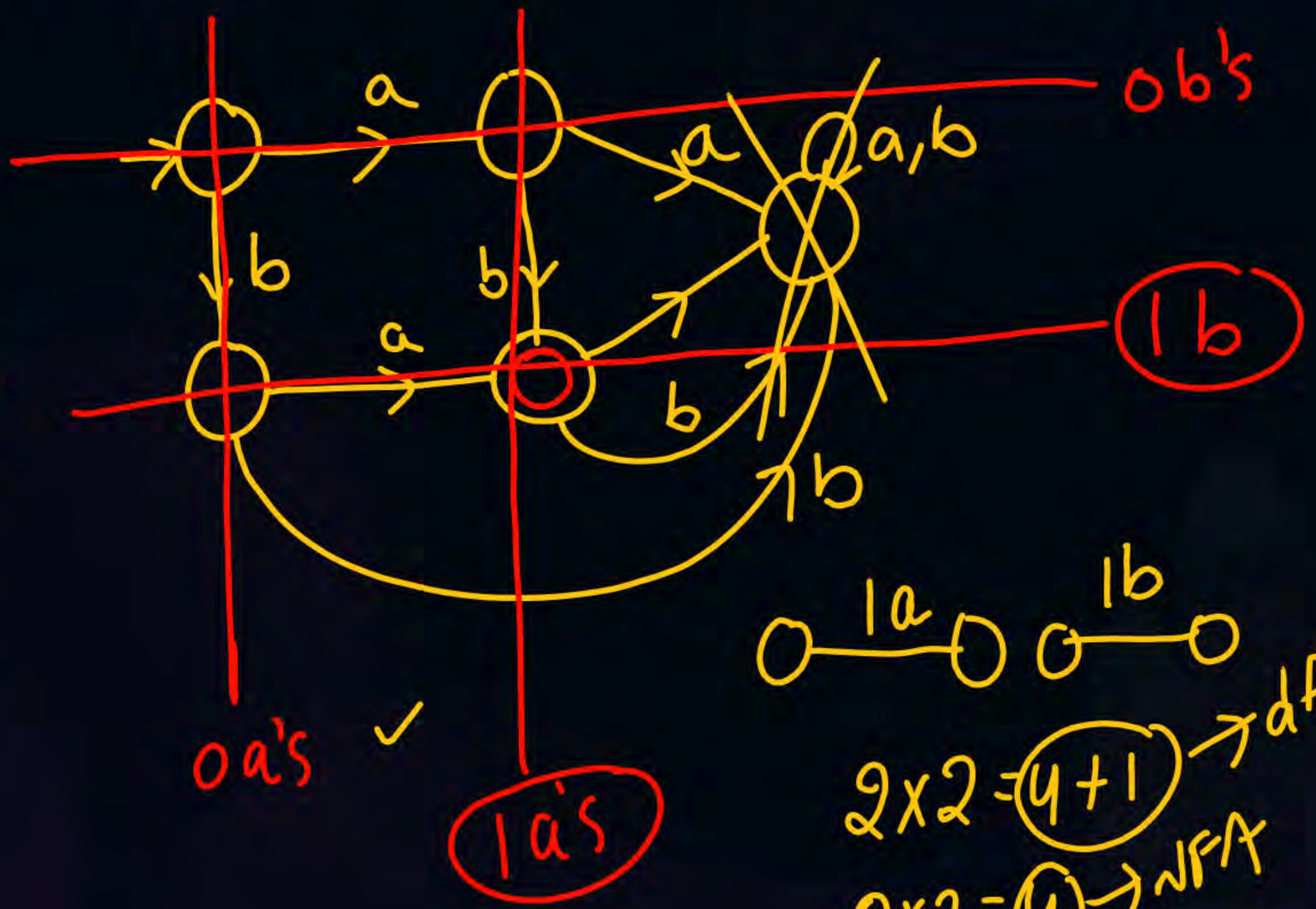
Exactly 2 a's



atmost 2 a's



$\Sigma = \{a, b\}$ $L = \{ w / w$ contains exactly 1a and exactly 1b $\}$

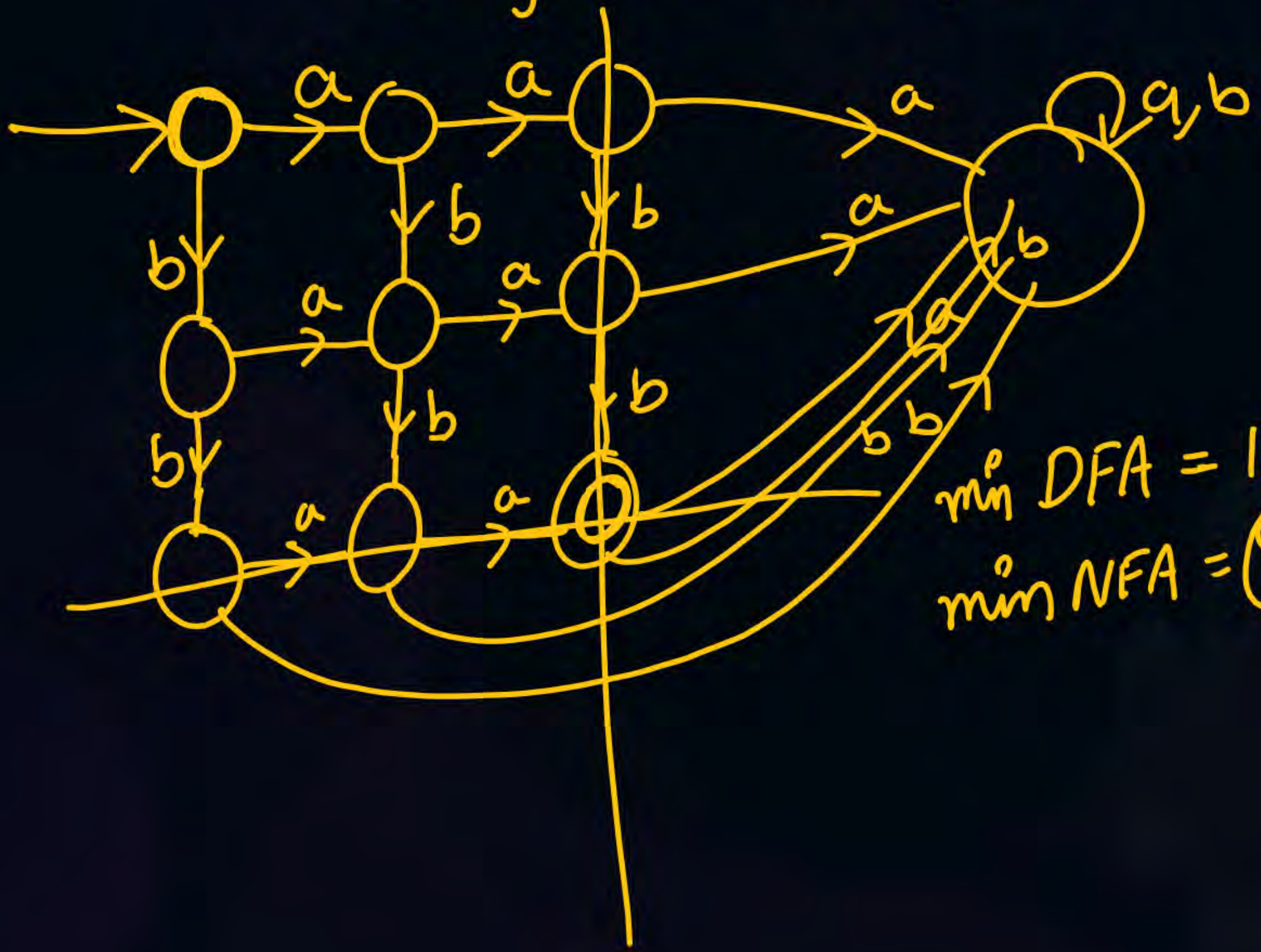


III - 86
IV - 13
mTech
NSC

$2 \times 2 = (4+1)$ → dfa

$2 \times 2 = (4)$ → NFA

exactly 2 a's exactly 2 b's



min DFA = 10

min NFA = 9

Exactly (2 a's) and exactly (3 b's)

min DFA =

min NFA =

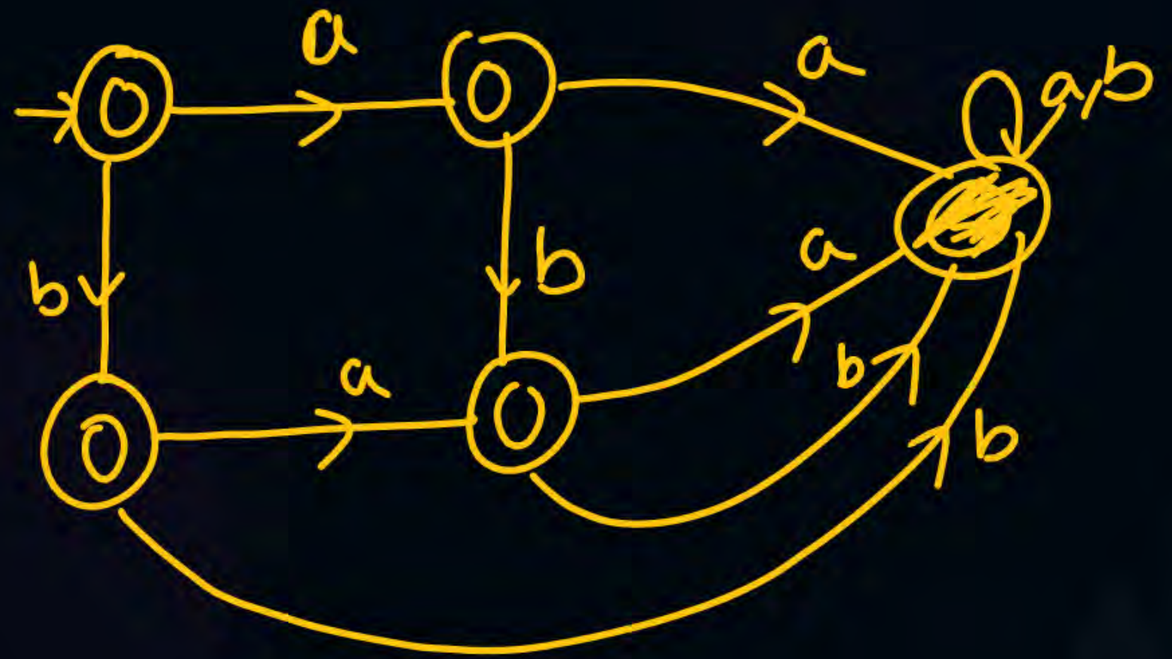
(2 a's) → 3 states

(3 b's) → 4 states

min DFA → (3 × 4) + 1 = 13

min NFA → (3 × 4) → 12

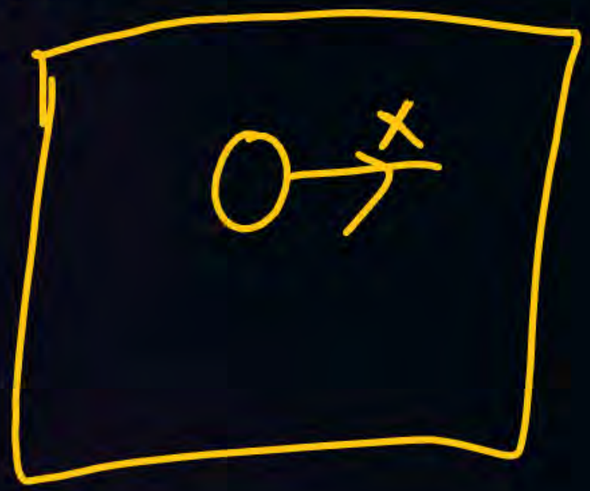
$\Sigma = \{a, b\}$ $L = \{w / w$ contains atmost 1a, atmost 1b$\}$



$\Sigma = \{a, b\}$ $L = \{w / w$ Contains atmost (5 a's) 6 atmost (5 b's) 6 $\}$

min DFA = 37 $\quad 6 \times 6 = 36$

min NFA = 36

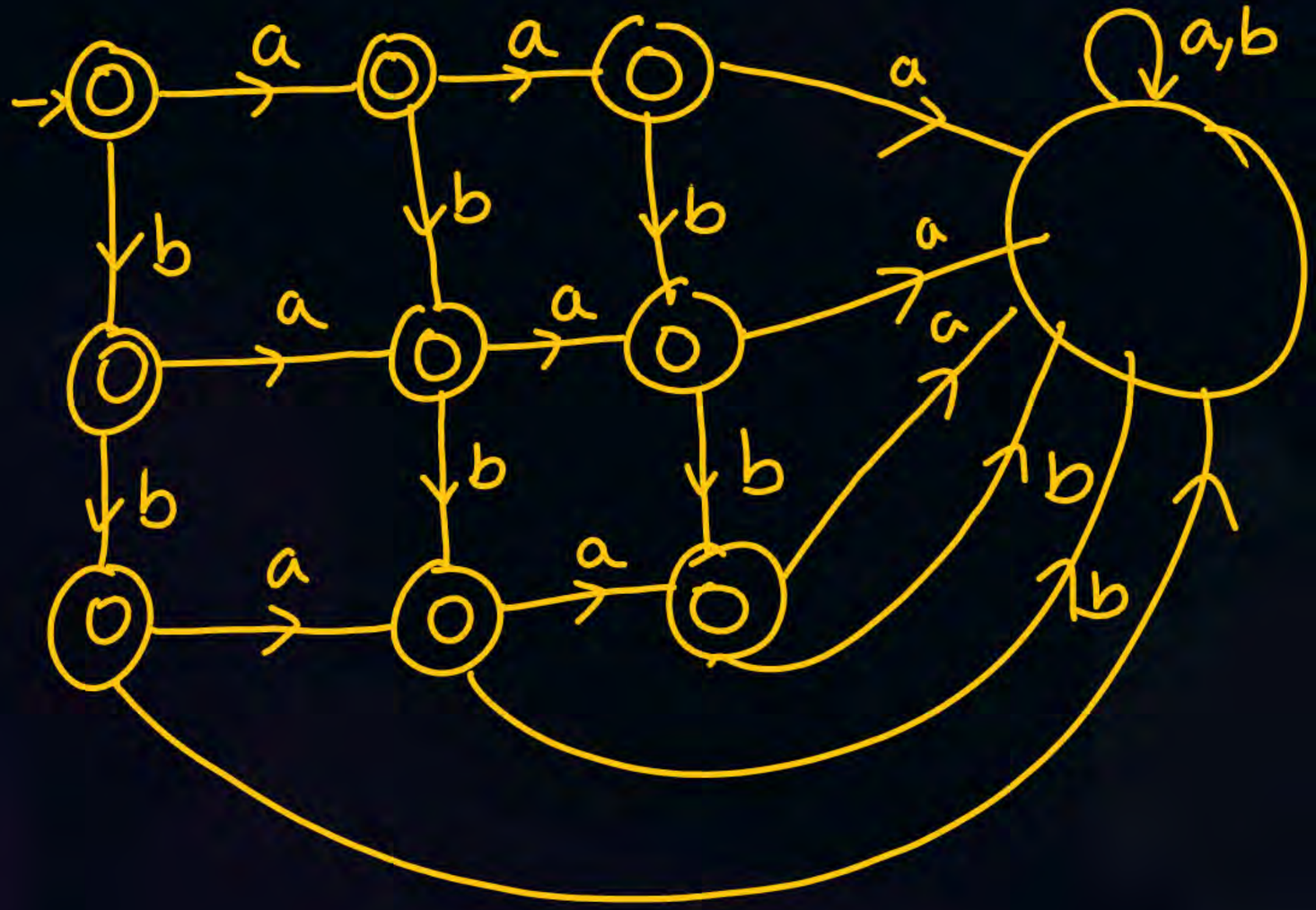




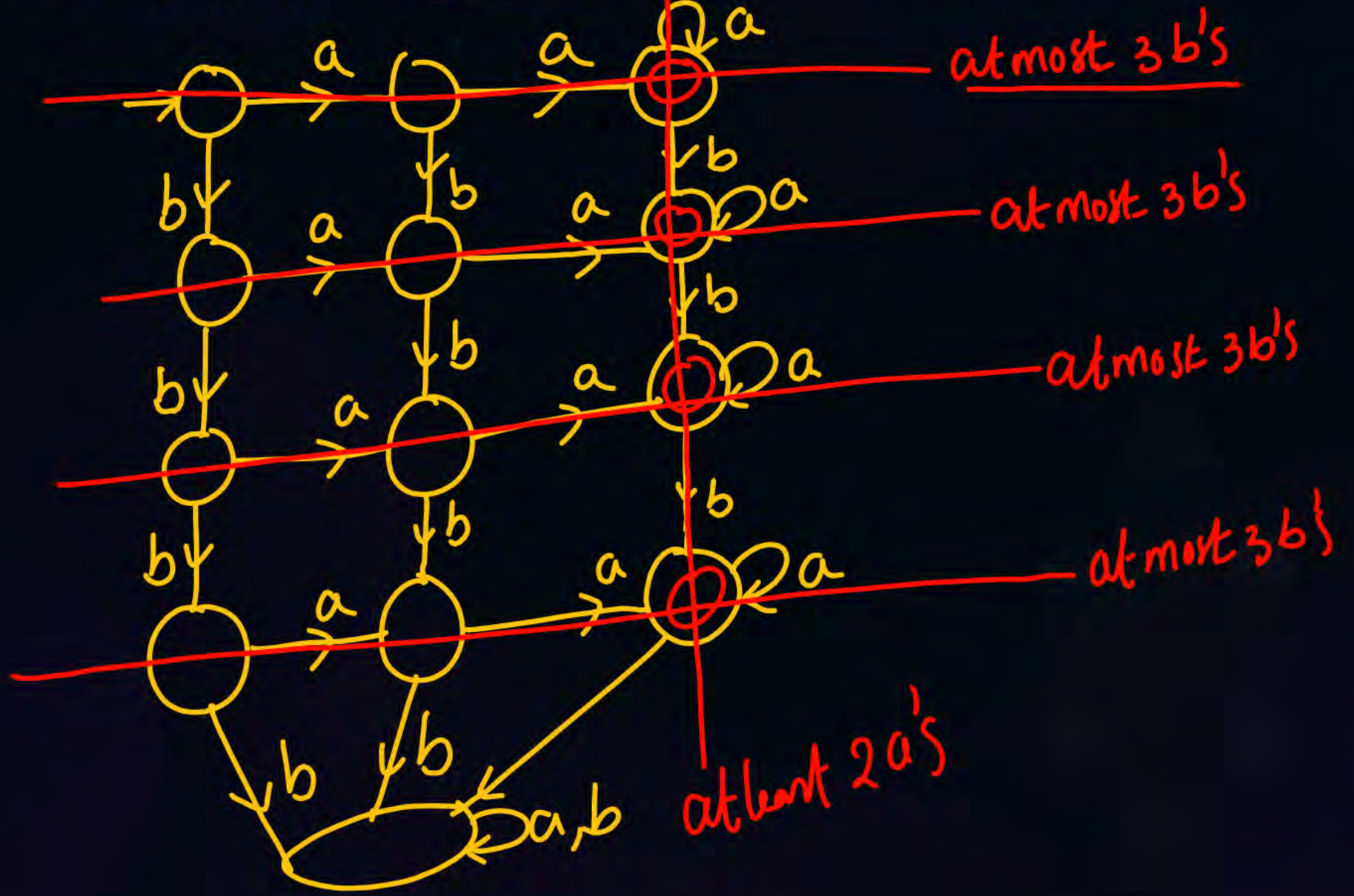
what is the language.

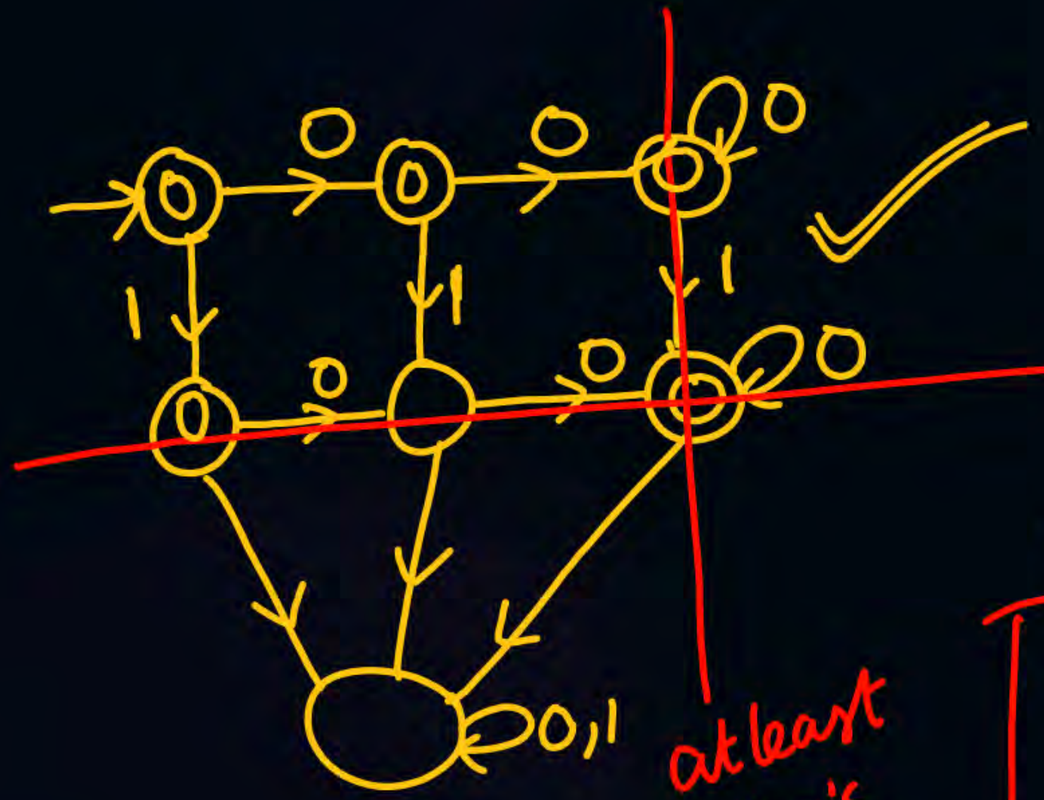
$\Sigma = \{a, b\}$ $L = \{w / w$ has atleast 2 as and atleast 2bs $\}$

$\Sigma = \{a, b\}$ at most 2 a's at most 2 b's



$\Sigma = \{a, b\}$ atleast 2 a's atmost 3 b's





exactly one 1

atleast 2 0's

atleast 2 0's and exactly one 1

atleast 3 a's and atmost 3 b's

min NFA 16

min DFA 17

4 X 4 =

$\Sigma = \{a, b, c\}$ ~~at least~~ atmost 2 a's, atleast 2 b's, atleast 2 c's



$3 \times 3 \times 3 = 27$

DFA $3 \times 3 \times 3 + 1$

NFA — 28, $3 \times 3 \times 3$, 27

Properties of RE

$r, s, t \rightarrow RE$

Commutative property

$$r + s = s + r$$

$$a^* + b^* = b^* + a^*$$

$$ab^* + ba^* = ba^* + ab^*$$

$r.s \neq s.r$ → may & may not

$ab \neq ba$

$\frac{a^*}{r} \cdot \frac{a}{s} = \frac{a}{s} \cdot \frac{a^*}{r}$

$a^*a = aa^* = a^+$

a^*a

$aa^* = a\{\epsilon, a, aa, aaa \cdots\cdots\}$

$= \{a, aa, aaa \cdots\} = a^+$

Associative :-

$$r+(s+t)=(r+s)+t$$

$$a^*+(b+c^*)=(a^*+b)+c^*$$

$$r(st)=(rs)t$$

$$a^*(bc^*)=(a^*b)c^*$$

$$aa^*=a^+ \qquad a^++\epsilon=a^*$$

$$aa^*+\epsilon=a^*$$

Distributive :-

$r.(s+t) = rs + rt$

$a^*(bc+d) = a^*bc + a^*d$

$r + (s.t) \neq (r+s).(r+t)$

$a + bc \neq (a+b)(a+c)$

$a + bc$ → a, bc

$(a+b)(a+c)$ → aa, ac, ba, bc

$+ \rightarrow$ union

$\cdot \rightarrow$ Concatenation

Identity:

$r + \phi = r = \phi + r$

$r \cdot \epsilon = \epsilon \cdot r = \epsilon \cdot r \cdot \epsilon = r$

$r \cdot \phi = \phi$

$r + \epsilon \neq r$ → may or may not

If r contains ϵ, then $r + \epsilon = r$

otherwise $r + \epsilon \neq r$

$a^+ + \epsilon \neq a^+$

$a^+ + \epsilon = a^*$

$\underline{a^*} + \epsilon = \underline{a^*}$

$\epsilon + \phi = \epsilon$

$\epsilon \cdot \phi = \phi$

$\phi^* = \epsilon$

$\phi^+ = \phi$

$\phi + \epsilon = \epsilon$ $\quad (a^* \cdot \phi)^*$ $\quad (\phi)^* = \epsilon$

$r + r = r$

$a^* + a^* = a^*$

$\underline{a} + \underline{a} = \underline{a}$

$r \cdot r \stackrel{?}{=} r$

$\underline{r} \cdot \underline{r} \neq r$ → may / may not

$a^* a^* = a^* \{\epsilon, a, aa, aaa \cdots\}$

$= \underline{a^*}, \underline{a^* a, a^* aa, \cdots}$

a^*

$\underline{a} \cdot \underline{a} \neq \underline{a}$

$\epsilon \cdot \epsilon = \epsilon$

$\underline{a^*} \cdot \underline{a^*} = \underline{a^*}$

$(r^*)^* = r^*$

$(a^*)^* = \{\epsilon, a, aa, aaa \ldots\}^*$

a^*

$a.a \; != a$

$\{aa\} \quad \{a\}$

$aa \neq a$

$a^*a^* = a^*$

$(a^{\oplus})^{\oplus'}$

a ✓ a^+

$(a^*)^{\oplus'} = a^*$ → everything

$(a^{\oplus})^* = a^*$ → everything

$(a^+)^+ = a^+$

$(a^+)^+ = \{a, aa, aaa, aaaa \ldots\}$ (+)

a^+ ✓

$a \oplus a = a$

$\{a\} \cup \{a\}$

$\{a\}$

$a^* \cdot a^+ = a^* \{a, aa, aaa \cdots\}$

$= a^*a, a^*aa, a^*aaa \cdots$

$= a\{\epsilon, a, aa, aaa \cdots\}a$

$= \{a, aa, aaa \cdots\}$

$= a^+$

$a^* \cdot a^+ = a^+ = a^+ \cdot a^* =$

° if $L(S) \subseteq L(r)$ then $r + S = r$

$a^* + aa^* = a^*$

$a^+ + aa = a^+$

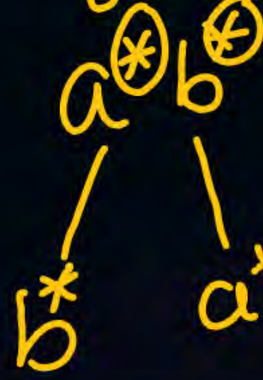
$(a+b)^*(a+ba)$

$(a+b)^*a + (a+b)^*ba$

$(a+b)^*a$



$a^*b^* + a^* = a^*(b^* + \epsilon) = a^*b^*$



$a^* \subseteq a^*b^*$

$b^* \subseteq a^*b^*$

$(a+b)^*a$

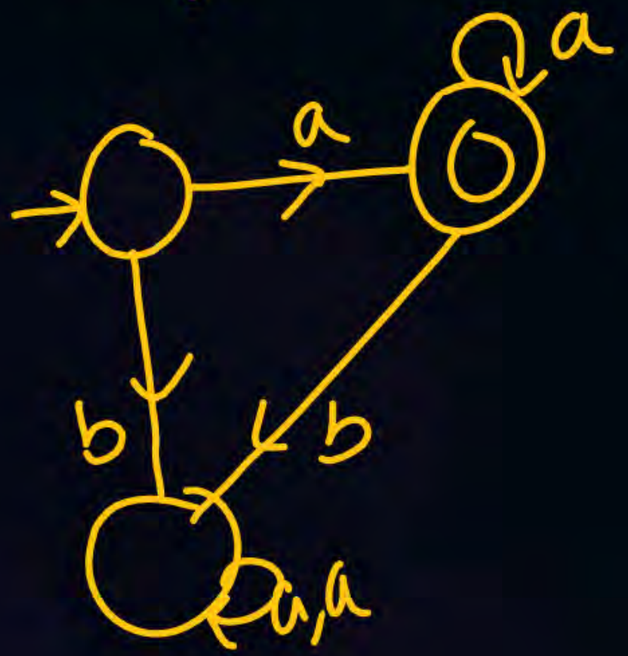
$(a+b)^*ba$

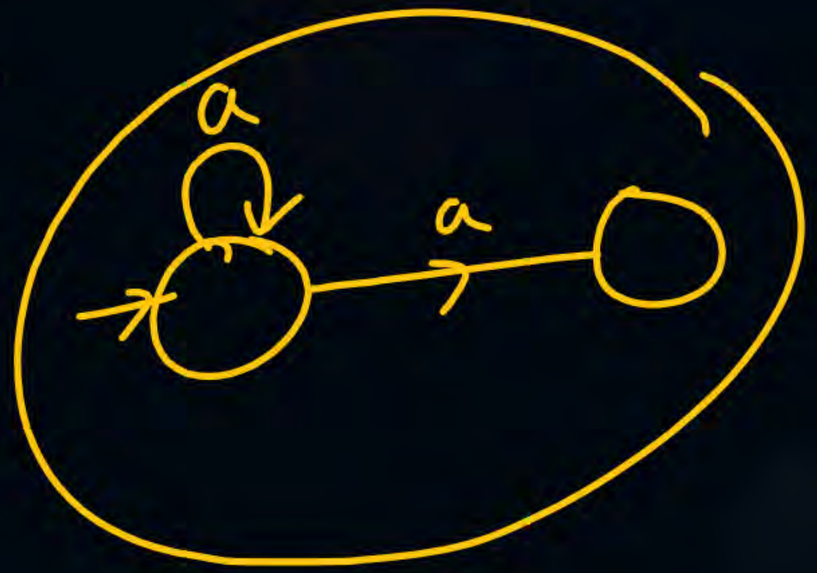
$(a+b)^*aa$

Subset

$\Sigma = \{a, b\}$

aa^* a^*a





$$\epsilon + a a^* = a^* = a^* a + \epsilon =$$

$$\epsilon + \gamma \gamma^*$$

$$\epsilon + (a^* b b^*)(a^* b b^*)^*$$

$$\epsilon + \gamma \gamma^*$$

$$= \gamma^*$$

$$(a^* b b^*)^*$$

$$(r+s)^* = (r^* + s^*)^* = (r^* s^*)^* = (s^* r^*)^*$$

$$= (r^* + s)^* = (r + s^*)^*$$

Find the odd one out

1) $(a^* b^*)^* \longrightarrow (a+b)^*$

2) $(b^* a^*)^* \longrightarrow (a+b)^*$

3) $(a^* + b^*)^* \longrightarrow (a+b)^*$

4) $(a^* + ba^*)^* \longrightarrow (a+b)^*$

5) $(a^* b + b^* b)^* \longrightarrow$ odd man

$(a^{*'} + ba^*)^*$

$(a+b)^*$

$\Sigma = \{a, b, c\}$

$(a^* b + b a + b c)^*$

$a \to b$

$b \to a$

$c \to c$

$\therefore (\ldots c)^*$

$=$

$(r+s)^* = (s^* r^*)^*$

$(s)\ r\ r\ s$

$s\,q\ s\,q\,q\ s\,q\,q\ s\,q$

THANK - YOU